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SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	syntax
Selected Domain	Domain 1
Selected SDG Track (1–17)	Main 4, Relevent 8
Project / Solution Name	CareerLens
Name (Leader)	'Ainaa Qistina Binti Mohd Nazrin
Email & Phone (Leader)	ainaaqistinanzrin@gmail.com | 01116220071
University Affiliation(s)	Universiti Teknikal Malaysia Melaka (UTeM)
Number of Team Members	2

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Artificial intelligence (AI) is transforming the global workforce, creating new career opportunities while changing the skills required for existing jobs. However, university students, fresh graduates, and working professionals often struggle to identify suitable career paths, understand emerging industry demands, and determine which skills they should develop to remain competitive.

According to the Hays Asia Salary Guide 2025, 64% of Malaysian organisations experienced moderate to extreme skills shortages, highlighting a significant gap between workforce capabilities and employer expectations.

Traditional job search platforms primarily focus on matching qualifications with current vacancies. They often provide limited explanations of missing skills, changing industry requirements, future career opportunities, and structured learning pathways.

Consequently, individuals may pursue unsuitable occupations, overlook emerging careers, or invest time and financial resources in training that does not align with evolving market demands.

This challenge particularly affects students preparing to enter the workforce, fresh graduates seeking employment, and working adults adapting to AI driven technological changes.

CareerLens aims to address this skills mismatch by providing personalised career recommendations, explainable skills gap analysis, emerging workforce insights, and practical reskilling guidance.

The solution supports SDG 4 (Quality Education) by promoting relevant lifelong learning and SDG 8 (Decent Work and Economic Growth) by helping individuals prepare for sustainable employment opportunities.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

CareerLens is an AI-powered career intelligence platform that helps individuals identify suitable future career opportunities, understand their skills gaps, and develop personalised learning pathways based on evolving job-market demands.

Users begin by uploading a CV or manually entering their skills, experiences, and interests. The system uses the Qwen3-8B AI model hosted through RunPod to extract and structure relevant information into a skills profile.

CareerLens employs an evidence based career matching engine that evaluates users' existing skills against structured occupational requirements obtained from sources such as O*NET. The algorithm considers software skill relevance, transferable skill coverage, and personal interests to produce explainable career fit scores and identify missing competencies. The platform also incorporates dated job market information and supports updates from external data sources, allowing recommendations to reflect changes in occupational information. AI generated explanations complement the deterministic matching process without independently inventing career opportunities.

The platform incorporates a Future Skills Radar that presents emerging and changing skill demands using dated, source attributed labour market information. It also compares moderate and accelerated AI disruption scenarios to help users understand potential career changes without treating predictions as guaranteed outcomes.

CareerLens then generates personalised reskilling pathways containing recommended learning activities, estimated durations, and accessible educational resources, including free tutorials and online courses.

Additionally, recommended occupations and skills are mapped to relevant United Nations Sustainable Development Goals, allowing users to explore how their career choices contribute to sustainable development.

The application uses React and TypeScript for its interface, Express for backend services, PostgreSQL for structured data management, Replit for development, and RunPod for AI inference.

CareerLens ultimately aims to transform conventional career searching into an informed, explainable, and continuous career development experience.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

CareerLens adopts a modular, web-based architecture integrating artificial intelligence, career matching algorithms, labour market intelligence, and persistent data storage.

The frontend is developed using React, TypeScript, Vite, and Tailwind CSS to provide a responsive user interface. The backend uses Node.js and Express to handle user profiles, CV processing, career recommendations, learning pathways, and trend analysis.

AI-powered operations are supported by the Qwen3-8B model through RunPod, enabling structured CV skills extraction and personalised career guidance. PostgreSQL, managed through Drizzle ORM, stores career information, profiles, skills, market signals, and learning roadmaps.

The career-matching engine uses source backed occupational information to calculate explainable career fit scores based on software skill relevance, transferable skill coverage, and user interests. Matching weights are adjusted according to the availability of supporting evidence, while job market signals provide additional context for career exploration.

The platform also includes functionality for importing occupation datasets and market trend observations, supporting future updates to career recommendations.

Replit serves as the primary development environment and intended application deployment platform.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

CareerLens follows a human centred design thinking approach consisting of five stages: Empathise, Define, Ideate, Prototype, and Test.

During the Empathise and Define stages, the team focuses on the challenges faced by students, fresh graduates, and working professionals in identifying relevant careers, understanding industry expectations, and preparing for AI driven workforce changes.

In the Ideate stage, the team develops solutions including AI powered CV analysis, explainable career matching, future skills monitoring, personalised learning pathways, and SDG integration.

During Prototyping, the team develops an interactive web application using React, TypeScript, Express, PostgreSQL, and RunPod. The initial implementation focuses on extracting user skills, generating career recommendations, identifying missing competencies, and presenting future workforce insights.

The testing stage incorporates automated tests for AI processing, career matching, occupation data handling, SDG associations, and market data updates. The team also plans to evaluate recommendation relevance and usability using synthetic profiles, while validating the consistency of skills gap results against source backed occupational requirements.

Before Phase 2, the team plans to refine the matching algorithm, improve dataset coverage, validate learning resources, and strengthen SDG explanations.

CareerLens will remain a decision support tool, allowing users to make informed career choices rather than relying on automated hiring decisions.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	**Your Response**
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	SDG 4.4: Increase relevant technical and vocational skills for employment and decent work. SDG 8.5: Promote productive employment and decent work for all.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	According to Hays Asia Salary Guide 2025, 64% of Malaysian organisations reported moderate to extreme skills shortages. This provides external evidence of the workforce skills mismatch CareerLens aims to address.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Percentage of synthetic user profiles receiving relevant source backed career recommendations, accurate skills gap identification, and actionable reskilling pathways.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Achieve at least 80% top three career recommendation relevance against a manually reviewed reference set, and provide source-supported learning pathways for at least 90% of successfully matched synthetic profiles.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- ☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- ☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- ☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- ☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☒ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
React	Code library	MIT
TypeScript	Code	Apache-2.0
Express	Code framework	MIT
PostgreSQL	Database software	PostgreSQL License
Drizzle ORM	Code library	Apache-2.0

Tailwind CSS	Code library	MIT
Qwen3-8B	AI model	Apache License 2.0
O*NET Database	Dataset	Creative Commons Attribution 4.0 (CC BY 4.0)
ESCO Skills Dataset	Dataset	ESCO Skills Pillar includes CC BY SA 4.0 material
World Economic Forum: Future of Jobs Report 2025	Dataset	WEF publication terms
Job Listings: Open Jobs in Every Field (Daily)	Dataset	Not verified (license information pending confirmation from Kaggle dataset publisher).
Future Jobs & Skills Demand 2025		CC0: Public Domain
Global AI Jobs 2026 Dataset		CC0: Public Domain

6.3 Employment-Related IP Conflicts

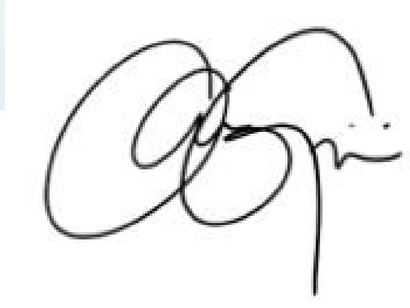
Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☒ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.



‘AINAA QISTINA BINTI MOHD NAZRIN

Date: 25/10/2026